The Effect of Different Exercise Protocols and Regression-Based Algorithms on the Assessment of the Anaerobic Threshold

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Abstract

Zuniga, JM, Housh, TJ, Camic, CL, Bergstrom, HC, Schmidt, RJ, and Johnson, GO. The effect of different exercise protocols and regression-based algorithms on the assessment of the anaerobic threshold. *J Strength Cond Res* 28(9): 2507–2512, 2014—The purpose of this study was to examine the effect of ramp and step incremental cycle ergometer tests on the assessment of the anaerobic threshold (AT) using 3 different computerized regression-based algorithms. Thirteen healthy adults (mean age and body mass [*SD*] = 23.4 [3.3] years and body mass = 71.7 [11.1] kg) visited the laboratory on separate occasions. Two-way repeated measures analyses of variance with appropriate follow-up procedures were used to analyze the data. The step protocol resulted in greater mean values across algorithms than the ramp protocol for the $\dot{V}O_2$ (step = 1.7 [0.6] $L\cdot min^{-1}$ and ramp = 1.5 [0.4] $L\cdot min^{-1}$) and heart rate (HR) (step = 133 [21] $b\cdot min^{-1}$ and ramp = 124 [15] $b\cdot min^{-1}$) at the AT. There were no significant mean differences, however, in power outputs at the AT between the step (115.2 [44.3] W) and the ramp (112.2 [31.2] W) protocols. Furthermore, there were no significant mean differences for $\dot{V}O_2$, HR, or power output across protocols among the 3 computerized regression-based algorithms used to estimate the AT. The current findings suggested that the protocol selection, but not the regression-based algorithms can affect the assessment of the $\dot{V}O_2$ and HR at the AT.

Key Words cycling, graded exercise testing, aerobic-anaerobic transition, ramp, step, protocol selection

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Introduction

Incremental cycle ergometer tests often differ in methodological characteristics, such as stage duration, size of the workload increment, and/or protocol used (10). These differences may be because of personal preference, the population studied, and/or type of cycle ergometer used (mechanically vs. electronically braked) (10,11).

Electronically braked cycle ergometers allow the resistance to be applied evenly, in a linear fashion, regardless of variations in pedaling speed (11). Previous investigations (11–13) have also indicated that large and unequal workload increments, such as those used during a step test protocol, resulted in inaccurate estimates of exercise capacity. Ramp protocols, however, have been shown to better estimate a number of physiological outcomes, such as the ability to predict oxygen uptake, a stronger relationship between exercise test time and ischemia, a wider distribution of time before the occurrence of ST-segment depression, and more uniform hemodynamic and gas exchange responses to exercise for patients with cardiovascular diseases (11). One potential limitation, however, may be the protocol (ramp vs. step) that is being used to assess physiological outcomes from an incremental cycle ergometer test.

One clinically significant physiological index that may be influenced by different testing protocols is the anaerobic threshold (AT). The AT is considered a useful assessment of functional capacity for cardiovascular and pulmonary patients (9) as well as a measurement of effectiveness of training adaptations in healthy individuals (8). A factor that limits the validity of the AT and makes comparisons difficult is the different methodologies and algorithms used to estimate the breaking point between the $\dot{V}CO_2$ vs. $\dot{V}O_2$ relationship (4). The determination of the AT based on visual inspection of the graphical representation of this relationship have been widely criticized for showing poor interevaluator and intermethod agreement (4). Most metabolic carts are packaged with algorithms that identify the AT, but it is unknown whether different protocols (ramp vs. step) result in different AT estimations.

To our knowledge, there are no studies that have compared the effect of ramp and step protocols on the $\dot{V}O_2$ and heart rate (HR) at the AT estimated using different computerized regression-based algorithms. This information would be important to know because many clinical settings rely on the estimated AT generated by the computer rather than by personnel. The purpose of this study, therefore, was to examine the effect of ramp and step incremental cycle ergometer tests on the assessment of the AT using 3 different computerized regression-based algorithms. Based on the findings of previous studies (4,17), we hypothesized that there would be significantly greater $\dot{V}O_2$, HR, and power output at the AT for the step than ramp protocol, as well as significant differences among the 3 different computerized regression-based algorithms.

Methods

Experimental Approach to the Problem

Each subject visited the laboratory on 3 separate occasions. The first visit was an orientation session. During the second and third visits, the subject performed a step or a ramp incremental cycle ergometer test to exhaustion in a random order. The data analyzed included $\dot{V}O_2$ and HR at the AT. The AT was assessed using 3 common computerized regression-based algorithms used in previous literature (4). The 3 mathematical models used were: (a) "Breaking point" algorithm developed by Jones and Molitoris (7), (b) "V-slope" algorithm developed by Beaver et al. (2), and (c) "D-max" algorithm developed by Cheng et al. (3). Repeated-measures analyses of variance (ANOVAs) were used to analyze the interaction and main effects between the protocol and algorithms.

Subjects

Inclusion criteria included healthy male or female 19–30 years of age who performed regular exercise at least 2 times per week. Exclusion criteria included body mass index of >30, lower extremity injury within the past 6 months, and medical conditions that would be contraindications to maximal exercise. The study was approved by the University Institutional Review Board for Human Subjects, and all the subjects completed a health history questionnaire and signed an informed consent document before testing.

Incremental Cycle Ergometer Tests

An orientation session was performed to familiarize the subjects with the maximal step and ramp incremental cycle ergometer tests that include pedaling at 70 rpm for 6 minutes up to 90 W. Twenty-four to 48 hours after the orientation session, each subject performed either a step or a ramp incremental cycle ergometer test to exhaustion in a random order on a Calibrated Lode (Lode, Gronigen, The Netherlands) electronically braked cycle ergometer at a pedal cadence of 70 rpm. Forty-eight hours after the second visit, subjects performed the other (step or ramp) cycle ergometer test. The testing sessions were performed at the same time of the day (±2 hours) with the same tester. In addition, subjects were encouraged to have an appropriate hydration and the last meal 2–3 hours before the step and ramp incremental cycle ergometer tests. The seat was adjusted so that the subject's legs were at near full extension during each pedal revolution. Seat height was recorded for each subject during the orientation visit and used for subsequent visits. The subject was fitted with a nose clip and breathed through a 2-way valve (2700; Hans Rudolph, Kansas City, MO, USA). Expired gas samples were collected and analyzed using a calibrated TrueMax 2400 metabolic measurement system (Parvo Medics, Sandy, UT, USA). Heart rate was monitored with a Polar Heart Watch system (Polar Electro, Inc., Lake Success, NY, USA). The Borg's 15-point rating of perceived exertion (RPE) was used during the step and ramp incremental cycle ergometer tests. The RPE was measured each minute of the tests using a RPE scale chart (Young Enterprises, Inc., Lansing, KS, USA). For the step incremental cycle ergometer test, the subjects started pedaling at 30 W, and power output was increased by 30 W every 2 minutes until voluntary exhaustion. For the ramp incremental cycle ergometer test, however, the resistance was increased progressively by 15 $W \cdot min^{-1}$ increments (1 W each 4 seconds). The $\dot{V}O_2$peak was defined as the highest $\dot{V}O_2$ value in the last 30 seconds of the test if the subject met at least 2 of the following 3 criteria: (a) 90% of age-predicted maximum HR (220-age), (b) respiratory exchange (RER) ratio >1.15, and (c) a plateau of oxygen uptake ($\leq$150 $ml \cdot min^{-1}$ in $\dot{V}O_2$ over the last 30 seconds of the test). At the completion of the test, subjects were allowed to cooldown. The test-retest reliability for $\dot{V}O_2$peak testing from our laboratory indicated the intraclass correlation coefficient was 0.95, with no significant mean difference between test and retest values.

Computer-Based Determination of the Anaerobic Threshold

A computer program (WinBrake 3.7, Ames, IA, USA) that incorporates the most common regression-based algorithms used to assess the AT derived from gas exchange parameters (4) was used in this study. A 20-second data averaging were used to be consistent with the methodology of previous investigations (4,5). The first method was the "Breaking point"

Table 1. Description of subjects.*

Variable	Women (n = 7) Mean (SD)	Men (n = 6) Mean (SD)
Age (y)	23.2 (2.9)	23.6 (4.2)
Body weight (kg)	63.1 (5.4)	80.4 (8.8)
Height (cm)	172.3 (5.4)	182.5 (4.7)
BMI ($kg \cdot m^{-2}$)	21.2 (1.6)	24.1 (1.6)

*BMI = body mass index.

Table 2. Mean and *SD* for the 3 regression-based algorithms to determine anaerobic threshold and the maximal values for the step and ramp cycling tests.*

Anaerobic threshold algorithm	Parameter: Ramp	Parameter: Step
Breaking point		
$\dot{V}O_2$ (L·min^{-1})	1.5 (0.4)	1.7 (0.6)
HR (b·min^{-1})	124 (15)	133 (21)
Power output (W)	112.2 (31.2)	115.2 (44.3)
V-slope		
$\dot{V}O_2$ (L·min^{-1})	1.4 (0.4)	1.7 (0.5)
HR (b·min^{-1})	121 (15)	132 (21)
Power output (W)	103.4 (25.7)	114.9 (42.3)
D-max		
$\dot{V}O_2$ (L·min^{-1})	1.6 (0.3)	1.7 (0.4)
HR (b·min^{-1})	126 (13)	136 (13)
Power output (W)	106.9 (38.6)	117.2 (26.7)

Maximal values	Parameter: Ramp	Parameter: Step	*p*
$\dot{V}O_2$peak (L·min^{-1})	3.0 (0.8)	2.9 (0.7)	0.42
HR max (b·min^{-1})	177 (8.7)	178 (10)	0.60
Peak power output (W)†	237.0 (47.3)	226.1 (53.4)	0.01
Time completed (min)†	15.9 (3.3)	14.8 (3.4)	0.01

*HR = heart rate.
†Significantly ($p \leq 0.05$) higher for the ramp than step cycling test.

algorithm developed by Jones and Molitoris (7). Briefly, this algorithm forces 2 regression equations, one before and one after the breaking point to join to identify the breaking point. The second method was the "V-slope" algorithm of Beaver et al. (2) that divides the $\dot{V}CO_2$ vs. $\dot{V}O_2$ relationship data points into 2 contiguous segments, fitting each segment with a linear regression line, and then calculating the intersection between the 2 regression lines. The third method was the "D-max" algorithm of Cheng et al. (3). This algorithm consists of fitting a third-order curvilinear regression to the $\dot{V}CO_2$ vs. $\dot{V}O_2$ relationship, drawing a straight line between the first and last data point of the curve and identifying the point that yields the maximal distance (D-max) from the straight line and third-order curvilinear regression line. All 3 algorithms were based on the $\dot{V}CO_2$ vs. $\dot{V}O_2$ relationship. Detailed descriptions about the 3 regression-based algorithms are presented elsewhere (2,3,7).

Statistical Analyses

Mean and *SD* (mean [*SD*]) were calculated for the 3 regression-based algorithms to determine the$\dot{V}O_2$, HR, and power output at the AT for the ramp and step cycling tests. Separate 2 (protocol: ramp and step) × 3 (algorithms: breaking point, V-slope, and D-max) repeated-measures ANOVAs were performed to analyze the $\dot{V}O_2$, HR, and power output associated with the AT. When appropriate, Tukey's Honest Significant Difference (HSD) post hoc testing was conducted.

Table 3. Mean and *SD* for main effects for protocol and algorithm.*

Parameter	Ramp	Step	*p*
Main effect for protocol (values collapsed across algorithm)			
$\dot{V}O_2$ (L·min^{-1})†	1.5 (0.3)	1.7 (0.4)	0.02
HR (b·min^{-1})†	120 (12)	133 (15)	0.01
Power output (W)	107.4 (26.7)	115.8 (31.4)	0.28

Parameter	Braking point	V-slope	D-max	*p*
Main effect for algorithm (values collapsed across protocol)				
$\dot{V}O_2$ (L·min^{-1})	1.6 (0.4)	1.5 (0.4)	1.6 (0.3)	0.82
HR (b·min^{-1})	129 (14)	126 (15)	131 (12)	0.95
Power output (W)	113.7 (30.3)	108.9 (29.7)	112.0 (25.8)	0.82

*HR = heart rate.
†Significantly ($p \leq 0.05$) higher for the step than ramp cycling test.

In addition, separate paired *t*-tests were used to compare $\dot{V}O_2$peak, maximum HR, peak power output, and time completed for the ramp vs. step cycling tests. A *p* value of $\leq$0.05 was considered statistically significant for all comparisons.

Results

Thirteen healthy adults with an age range form 20 to 28 years of age volunteered to participate in the investigation. The subjects participated regularly in walking ($n = 2$), bicycling ($n = 3$), jogging or running ($n = 11$), weight training ($n = 9$), and swimming ($n = 4$). Table 1 shows the demographic data of the subjects.

Table 3 shows the main effects for protocol and algorithm. There were no significant protocol × algorithms interactions for the $\dot{V}O_2$ ($F_{(2,24)} = 0.19$; $p = 0.82$), HR ($F_{(2,24)} = 0.05$; $p = 0.95$), or power output values ($F_{(2,24)} = 0.18$; $p = 0.82$) associated with the AT. There was a significant main effect, however, for protocol for $\dot{V}O_2$ and HR at the AT. The step cycling test resulted in significantly higher $\dot{V}O_2$ and HR values collapsed across algorithms than the ramp (Table 3). There were no significant mean differences, however, in power outputs at the AT values collapsed across algorithms between the step and the ramp protocols (Table 3).

Discussion

A major finding of this study was that the step cycling test resulted in significantly greater $\dot{V}O_2$ and HR at the AT than the ramp cycling test with no differences in power output. There were no mean differences, however, among the 3 computerized regression-based algorithms for the$\dot{V}O_2$, HR, or power output at the AT. In agreement with the results of this study, previous investigations (1,17) have shown differences between ramp and step cycling protocols on gas exchange parameters at the AT, as well as the overall metabolic responses, such as $\dot{V}O_2$ and HR.

Amann et al. (1) examined the influence of 2 different incremental cycling tests protocols with different stage durations and workload increments on 5 different methods to assess the AT, such as (a) RER = 1.0, (b) RER = 0.95, (c) ventilatory equivalents = $V_E/\dot{V}O_2$, (d) V-slope method, and (e) breaking point of the $V_E/\dot{V}CO_2$ relationship. It was found (1) from the 5 different methods to assess the AT, only the ventilatory equivalent method ($V_E/\dot{V}O_2$) resulted in significantly different $\dot{V}O_2$ at the AT. Based on the significantly greater minute ventilation (i.e., V_E), Amann et al. (1) concluded that different power increments and stage durations between protocols influenced the $V_E/\dot{V}O_2$ relationship resulting in a different $\dot{V}O_2$ at the AT. Amann (1) speculated that the changes in the $V_E/\dot{V}O_2$ relationship between protocols may have been because of the differences in the feedback from central and peripheral chemoreceptors, such as increases in partial pressure of CO_2, hydrogen ions, potassium, catecholamines, and body temperature which are the factors known to stimulate gas exchange parameters. Specifically, it was suggested (1) that muscular fatigue resulting from greater power increments and stage durations may have had an effect on the central and peripheral sensory control mechanisms of V_E.

The effect of different cycling protocols with different distribution of the workload has been also shown to affect the $\dot{V}O_2$ vs. power output relationship (17). It has been proposed (17) that the protocol selection, such as ramp and step cycling tests, can affect the $\dot{V}O_2$ vs. power output relationship. Zuniga et al. (17) found that differences in the distribution of the resistance with the same workload increments for the 2-minutes stage (ramp = 15 $W \cdot min^{-1}$ vs. step = 30 W every 2 minutes) resulted in lower submaximal mean values for $\dot{V}O_2$ and HR during the ramp than the step cycling test. It was concluded (17) that the cumulative effect of producing a greater amount of work during the step (75.83 kJ) than the ramp (65.60 kJ) cycling tests may have resulted in a greater $\dot{V}O_2$ and HR responses. Thus, based on the results of previous findings (1,17) it is conceivable that the lower $\dot{V}O_2$ and HR at the AT for the ramp ($\dot{V}O_2$ = 1.5 [0.4] $L \cdot min^{-1}$ and HR = 124 [15] $b \cdot min^{-1}$) than the step ($\dot{V}O_2$ = 1.7 [0.6] $L \cdot min^{-1}$ and HR = 133 [21] $b \cdot min^{-1}$; Table 2) cycling tests found in this study may be because of the smaller and progressive increases in power output for the ramp than step cycling test (1), as well as the difference in work performed (17). It is likely that muscular fatigue resulting from greater power output increments (1) and greater amount of work (17) for the step than ramp cycling test may have resulted in the significantly lower peak power output (step = 226.1 [53.4] W and ramp = 237.0 [47.3] W; Table 2) and time completed (step = 14.8 [3.4] minute and ramp = 15.9 [3.3] minutes) found in this study for the step than the ramp cycling tests.

The results from this study and those from previous investigations (1,17) are in accordance with the observations made by Myers and Bellin (11) that $\dot{V}O_2$ is overestimated from tests that contain large increments in work rate, such as those from step cycling protocols. In contrast to the results of this study, previous investigations have found no significantly differences between ramp and step cycling tests for gas exchange parameters at the AT (6,11–13,16). Whipp et al. (16) was the first study to examine the effect of ramp vs. step incremental cycle ergometer test and found no significant differences for the $\dot{V}O_2$ at the AT. Myers et al. (12) and Glaser et al. (6) also found no differences between ramp and step tests on several hemodynamic and gas exchange parameters at the AT. The reason for the conflicting findings from this study and those from previous investigations (6,11–13,16) is unclear, but it can be speculated that factors related to the difference in workload increments for the ramp and step protocols or the method used for the determination of the gas exchange parameters at the AT (computerized vs. visual inspection) may have contributed to the conflicting results. Specifically, this investigation used a ramp cycling protocol with increases of 15 $W \cdot min^{-1}$ (or 1 W every 4 seconds) and a step cycling protocol with increases of 30 W every 2 minutes, and thus the overall increment for a 2-minute stage was the same regardless of the protocol. Among the studies that have reported no significant differences between ramp and step tests on gas exchange parameters at the AT (6,11–13,16), only Revill et al. (13) used a similar study methodology to that of this study. Although there were no significant differences between ramp and step cycling test on several gas exchange parameters, Revill et al. (13) indicated that the continuous changing work rate profile of the ramp protocol was perceived to be less demanding and may be more readily tolerated by patients with exertional dyspnea. Furthermore, all of those studies (6,11–13,16) that conflict with the results of this investigation used a visual inspection method for the assessment of the AT. It has been suggested (4) that the determination of the AT based on visual inspection of data plots resulted in poor interevaluator and intermethod agreement. Computerized methods, such as those used in this study, however, offer a more objective and reliable assessment of AT (4).

This investigation found that there were no mean differences among the 3 computerized regression-based algorithms for the $\dot{V}O_2$, HR, and power output at the AT. Previous investigations (4,15) have indicated that the main factor that limits the practicality of the determination of the AT from gas exchange parameters is the different methodologies and mathematical algorithms used to estimate the breaking point of the data of interest. Ekkekakis et al. (4) found that different regression-based computerized methods for determining the AT using gas exchange parameters can yield substantially different results. Furthermore, Ekkekakis et al. (4) indicated that the differences among regression-based computerized methods were large enough (0.2–19% of the $\dot{V}O_2max$) to raise concerns for both research and clinical practice. The result of this study, however, found that the $\dot{V}O_2$, HR, or power output at the AT for the breaking point (7), V-slope (2), and D-max (3) regression-based algorithms were not significantly different (Table 3). In accordance with the results of this investigation, Santos and Giannella-Neto (14) reported no significant differences among 5 different computerized methods used for the assessment of the AT. Specifically, Santos and Giannella-Neto (14) found no significant differences on the assessment of the AT among the RER (79 [8] %$\dot{V}O_2max$), end-tidal PO_2 (P_{ET,O_2}; 73 [11] %$\dot{V}O_2max$), ventilatory equivalent ($V_E/\dot{V}O_2$; 76 [10] %$\dot{V}O_2max$), pulmonary ventilation (V_E; 75 [9] %$\dot{V}O_2max$), and the V-slope method ($\dot{V}CO_2/\dot{V}O_2$; 83 [6] % $\dot{V}O_2max$) measured as a percentage of the $\dot{V}O_2max$. Thus, the findings of this study expand those of Santos and Giannella-Neto (14) suggesting that the equality among different methods for the assessment of the AT also applies to different regression-based algorithms (i.e., breaking point, V-slope, and D-max algorithms).

Practical Applications

The current findings indicated that the protocol selection (ramp vs. step) affected the assessment of the AT. Protocol selection should be considered when assessment and comparisons of HR or $\dot{V}O_2$ at the AT are performed. Specifically, the significantly greater $\dot{V}O_2$ and HR at the AT found for the step than the ramp cycling test with no differences in power output indicated that cardiovascular and metabolic variables may be overestimated from step cycling protocols. Thus, it is recommended that if a step protocol is used, the increment in power output should be less than 30 W. If increments in power output larger than 30 W are needed, the use of a ramp protocol is recommended. Furthermore, the results of this study expand those from Santos and Giannella-Neto (14) suggesting that the equality among different methods for the assessment of the AT also applies to different regression-based algorithms. Thus, based on the results of this study it is suggested that the protocol selection but not the regression-based algorithms can affect the assessment of the $\dot{V}O_2$ and HR at the AT. This information is important because many athletes, coaches, and strength and conditioning specialists use different exercise protocols for the assessment of the AT. Future studies should examine the effect of protocol selection on neuromuscular fatigue parameters to assess the influence of the workload distribution and size of the increment on muscular fatigue.

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